

Literature-Implied Full Negative Answer: Positive Pettis Integrability Need Not Be Strong

Run `fa_banach_001`, agent `agent_lane_08`

9 August 2026

Status

Status: **literature-implied full negative answer** to Problem 4.7 and to the second question in Problem 4.8 of arXiv:1812.00597. The decisive example is Example 1.12 of Musiał (2011); the implication to strong Pettis integrability below appears not to be stated there and is identified directly in this packet.

Source questions

Candeloro, Di Piazza, Musiał, and Sambucini ask on PDF page 11 of [1]:

Problem 4.7. Is each positive Pettis integrable multifunction McShane integrable?

Problem 4.8, second question. Is each positive Pettis integrable multifunction strongly Pettis integrable?

Here a Pettis-integrable multifunction $\Gamma: [0, 1] \rightarrow \text{cb}(X)$ is *strongly Pettis integrable* when its Pettis primitive M_Γ is countably additive in the Hausdorff metric, that is, when it is an h -multimeasure. The same paper's Theorem 4.2 (PDF pages 9–10) states that a $\text{cb}(X)$ -valued multifunction is McShane integrable if and only if it is both strongly Pettis and Henstock integrable [1].

Supporting example

Example 1.12 of Musiał [2, p. 779; PDF p. 11] chooses a scalarly integrable $f: [0, 1] \rightarrow c_0$ such that

$$T_f: c_0^* = \ell_1 \longrightarrow L_1[0, 1], \quad T_f(x^*) = \langle x^*, f(\cdot) \rangle,$$

is not weakly compact, and defines

$$\Gamma(t) = \text{conv}\{0, f(t)\}.$$

The example records that Γ is $\text{ck}(c_0)$ -valued, scalarly integrable, and Pettis integrable in $\text{cb}(c_0)$. Moreover

$$s(x^*, \Gamma(t)) = \max\{0, \langle x^*, f(t) \rangle\} = \langle x^*, f(t) \rangle^+ \geq 0, \tag{1}$$

so Γ is positive in precisely the sense used in the source question.

Identification

Proposition 1. *The multifunction Γ in Musiał's Example 1.12 is Pettis integrable but not strongly Pettis integrable. Consequently it is not McShane integrable.*

Proof. Let M_Γ be the $\text{cb}(c_0)$ -valued Pettis primitive of Γ . Assume for contradiction that it is an h -multimeasure. Use the Rådström support-function embedding

$$j: \text{cb}(c_0) \longrightarrow \ell_\infty(B_{\ell_1}), \quad j(C)(x^*) = s(x^*, C).$$

This is an isometry from Hausdorff distance to the supremum norm and converts Minkowski addition into vector addition. Hence

$$m(E) := j(M_\Gamma(E))$$

is a norm-countably additive $\ell_\infty(B_{\ell_1})$ -valued vector measure. In particular, if $E_n \downarrow \emptyset$, continuity of a vector measure from above gives

$$\|m(E_n)\|_\infty \longrightarrow 0. \tag{2}$$

The defining identity for the Pettis primitive and (1) give, for every measurable E and $x^* \in B_{\ell_1}$,

$$m(E)(x^*) = s(x^*, M_\Gamma(E)) = \int_E \langle x^*, f(t) \rangle^+ dt.$$

All these coordinates are nonnegative. Therefore

$$\begin{aligned} \sup_{x^* \in B_{\ell_1}} \int_E |\langle x^*, f(t) \rangle| dt &\leq \sup_{x^* \in B_{\ell_1}} \int_E \langle x^*, f(t) \rangle^+ dt \\ &\quad + \sup_{x^* \in B_{\ell_1}} \int_E \langle -x^*, f(t) \rangle^+ dt \\ &\leq 2\|m(E)\|_\infty. \end{aligned}$$

Taking $E = [0, 1]$ proves L_1 -boundedness of $T_f(B_{\ell_1})$; taking $E = E_n$ and using (2) proves its uniform countable additivity. Equivalently, $T_f(B_{\ell_1})$ is uniformly integrable. The Dunford–Pettis weak-compactness criterion in L_1 now implies that $T_f(B_{\ell_1})$ is relatively weakly compact. Thus T_f is weakly compact, contrary to the defining property in Example 1.12. This contradiction proves that M_Γ is not an h -multimeasure.

Finally, Theorem 4.2 of [1] implies that every McShane-integrable $\text{cb}(X)$ -valued multifunction is strongly Pettis integrable. Hence this Γ cannot be McShane integrable. $\square$

Independent confirmation of the McShane answer

Example 2.10 of Candeloro, Di Piazza, Musiał, and Sambucini [3, PDF p. 9] independently starts with a Pettis-integrable but non-McShane-integrable vector function g and sets $G(t) = \text{conv}\{0, g(t)\}$. It explicitly concludes that G is a $\text{ck}(X)$ -valued Pettis-integrable multifunction containing 0 everywhere, hence positive, but is neither Henstock nor McShane integrable. Thus the negative answer to Problem 4.7 is also present in the later literature without using the strong-Pettis implication above.

Provenance and scope

The construction and the non-weak-compactness of T_f are published in Musiał's 2011 Example 1.12. The supporting paper predates the source questions and does not state the strong-Pettis consequence. This packet is therefore classified as an *agent-identified literature implication*, not as a new counterexample produced by the run.

The result fully answers Problem 4.7 and the second question of Problem 4.8. It does *not* answer Problem 4.8's first question about positive Henstock-integrable multifunctions possessing Henstock-integrable selections. Nor does it answer the stronger question, posed as Question 2.14 of [3], asking for a positive Henstock- and Pettis-integrable $\text{cb}(c_0)$ -valued multifunction that is not strongly Pettis integrable. Nothing cited here shows that Musiał's Example 1.12 is Henstock integrable.

Review focus

The only non-verbatim bridge needing expert verification is the passage from Hausdorff countable additivity of M_Γ to uniform integrability of $T_f(B_{\ell_1})$. The displayed calculation isolates that bridge: the Rådström image is a norm-countably additive vector measure, and positivity turns its norm continuity into uniform control of both positive and negative parts of every scalar evaluation.

References

- [1] D. Candeloro, L. Di Piazza, K. Musiał, and A. R. Sambucini, *Integration of multifunctions with closed convex values in arbitrary Banach spaces*, arXiv:1812.00597v4 (2019), <https://arxiv.org/abs/1812.00597>.
- [2] K. Musiał, *Pettis Integrability of Multifunctions with Values in Arbitrary Banach Spaces*, Journal of Convex Analysis 18 (2011), no. 3, 769–810, https://www.heldermann-verlag.de/jca/jca18/jca1004_b.pdf.
- [3] D. Candeloro, L. Di Piazza, K. Musiał, and A. R. Sambucini, *Multifunctions determined by integrable functions*, arXiv:1906.07019v2 (2020), <https://arxiv.org/abs/1906.07019>.